

## 02 Architecture

### GCP Data Engineering - Healthcare Revenue Cycle Management (RCM)

This project focuses on building a data lake in Google Cloud Platform (GCP) for Revenue Cycle Management (RCM) in the healthcare domain.

#### GCP Services Used:

- Google Cloud Storage (GCS)
- BigQuery
- Dataproc
- Cloud Composer
- Cloud SQL (with MySQL)
- GitHub
- Cloud Build
- CI/CD

#### Techniques Involved:

- Metadata Driven Approach
- SCD Type 2 implementation
- Common Data Model (CDM)
- Medallion Architecture
- Logging and Monitoring
- Error Handling
- Optimizations
- CI/CD implementation
- Many more best practices

#### Data Sources:

- **EMR Data from 2 hospitals (Cloud SQL database):** Stores patients, providers, departments, encounters, and transactions from two hospitals.
- **Claims Data (Flat Files in GCS):** Monthly insurance claims data ingestion.
- **External APIs:** Fetching CPT, NPI, and ICD codes for standardization.

